

Seminar handout

Goodwillie Calculus Seminar

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1 Introduction

In homotopy theory, Goodwillie calculus provides a powerful framework for understanding functors between spaces or spectra. It draws a profound analogy with classical differential calculus: just as smooth functions can be approximated by Taylor polynomials, homotopy functors can be approximated by "polynomial functors" of increasing degree. The core of this theory decomposes a functor $F : \mathcal{S} \rightarrow \mathcal{S}$ into a tower of approximations, the Taylor tower, whose layers (the homogeneous functors) are classified by spectra with actions of the symmetric groups. This has led to deep insights into stable homotopy theory and manifold calculus.

property 1.1 Goodwillie Calculus. For a functor $F : \mathcal{S} \rightarrow \mathcal{S}$, Goodwillie calculus constructs a Taylor tower $\cdots \rightarrow T_n F \rightarrow T_{n-1} F \rightarrow \cdots \rightarrow T_0 F$ to approximate F . Its layers $D_n F = \text{hofib}(T_n F \rightarrow T_{n-1} F)$ are homogeneous functors of degree n , classified by spectra equipped with an action of Σ_n .

Orthogonal calculus, introduced by Michael Weiss in 1995, is a close relative of Goodwillie's theory but is designed for a different, geometrically rich setting. It primarily focuses on functors that possess an inherent orthogonal symmetry.

Definition 1.1 Category $\mathcal{J}$. The domain category for orthogonal calculus is the category $\mathcal{J}$, where:

- **Objects** are finite-dimensional real inner-product spaces.
- **Morphisms** are linear isometric embeddings.

Formally, we study continuous functors $F : \mathcal{J} \rightarrow \mathcal{S}$ (taking values in spaces or spectra). The motivation stems from fundamental objects in topology and geometric topology that naturally carry such a structure.

Example 1.1 Classical Functors with Orthogonal Symmetry. Common examples of functors $F : \mathcal{J} \rightarrow \mathcal{S}$ include:

- $V \mapsto BO(V)$, the classifying space of the orthogonal group of V .
- $V \mapsto B\text{Top}(V)$, the classifying space of the homeomorphism group of V .
- $V \mapsto \text{Emb}(M, V)$, the space of embeddings of a fixed compact manifold M into V .

Each of these can be viewed as a single "large" space (such as BO , $B\text{TOP}$, or $\text{Emb}(M, \mathbb{R}^\infty)$) filtered by the finite-dimensional subspaces $V \subset \mathbb{R}^\infty$.

The central problem orthogonal calculus addresses is: How can we systematically analyze and approximate such functors to understand their behavior as the dimension of V grows?

property 1.2 Taylor Theory of Orthogonal Calculus. Orthogonal calculus constructs a Taylor tower of polynomial approximations $T_n F(V)$ for a functor $F : \mathcal{J} \rightarrow \mathcal{S}$. Its derivatives (the

building blocks) are not merely spectra but spectra equipped with canonical actions of the orthogonal groups $O(n)$, reflecting the enhanced symmetry present in the domain category $\mathcal{J}$. This framework has proven highly effective, for instance, in computing the rational homology of embedding spaces, demonstrating that orthogonal calculus is not just an abstract theory but a potent computational tool in geometric topology.

2 Categories and the Definition of Polynomiality

The foundation of orthogonal calculus is built upon specific topological categories that encode the notion of "direction" and "jet" data.

Definition 2.1 Base Category $\mathcal{J}$. The base category $\mathcal{J}$ has as objects finite-dimensional real inner-product spaces. The morphism space $\text{mor}(V, W)$ is the Stiefel manifold of linear isometric embeddings from V to W . This gives $\mathcal{J}$ the structure of a topological category.

To capture higher-order differential information, Weiss introduces an enhanced sequence of categories $\mathcal{J}_n$.

Definition 2.2 Jet Categories $\mathcal{J}_n$. For each $n \geq 0$, the category $\mathcal{J}_n$ has the same objects as $\mathcal{J}$. The morphism space $\text{mor}_n(V, W)$ is defined as the Thom space of a specific vector bundle $\gamma_n(V, W)$ over the Stiefel manifold $\text{mor}(V, W)$. The fiber of this bundle over a map $f : V \hookrightarrow W$ is $n \cdot \text{coker}(f)$, i.e., $\mathbb{R}^n \otimes (W/f(V))$.

Intuitively, a morphism in $\mathcal{J}_n$ is a pair (f, x) where f is an embedding and x is an element in the n -fold tensor product of the cokernel, considered "up to homotopy" at the Thom space level. The composition law is given by $((f, x), (g, y)) \mapsto (f \circ g, x + f_*(y))$.

There are natural inclusions $\mathcal{J}_0 \subset \mathcal{J}_1 \subset \mathcal{J}_2 \subset \dots$, where $\mathcal{J}_0$ is essentially $\mathcal{J}$ with a disjoint basepoint added. One should think of $\mathcal{J}_n$ as the " n -th jet category" of $\mathcal{J}_0$.

We study continuous functors from these categories to pointed spaces.

Definition 2.3 Category of Continuous Functors $\mathcal{E}_n$. Let $\mathcal{E}_n = \text{Fun}_{\text{cont}}(\mathcal{J}_n, \mathcal{S}_*)$ denote the category of continuous functors from $\mathcal{J}_n$ to pointed spaces. Continuity means that for every V, W , the evaluation map

$$\text{mor}_n(V, W) \wedge E(V) \rightarrow E(W)$$

is continuous. A functor in $\mathcal{E}_0$ is our primary object of study.

Definition 2.4 Polynomial Functor (Definition 5.1). A continuous functor $F : \mathcal{J} \rightarrow \mathcal{S}_*$ is said to be polynomial of degree $\leq n$ if, for every inner-product space V , the canonical map

$$\rho : F(V) \longrightarrow \text{holim}_{0 \neq U \subset \mathbb{R}^{n+1}} F(U \oplus V)$$

is a homotopy equivalence. Here, the homotopy limit is taken over the topological poset of all nonzero linear subspaces U of $\mathbb{R}^{n+1}$, ordered by inclusion.

property 2.1 Geometric Interpretation and Comparison. The geometric meaning of this definition is one of extrapolation. The homotopy limit $\text{holim}_U F(U \oplus V)$ represents the best

possible reconstruction of $F(V)$ if one only knows the values $F(U \oplus V)$ for all nontrivial extensions of V by a subspace of $\mathbb{R}^{n+1}$. A functor is polynomial if this reconstruction is perfect; knowing F on all such "thickened" spaces completely determines $F(V)$.

This is conceptually different from the definition in Goodwillie's calculus. In Goodwillie's setup, a functor is polynomial of degree $\leq n$ if it sends certain strongly cocartesian $(n+1)$ -cubical diagrams to cartesian diagrams—a condition rooted in excision. The orthogonal definition replaces cubical diagrams with this homotopy limit over the Grassmannian of directions, which is more natural for functors depending on a vector space variable.

3 Derivatives and Derivatives-at-Infinity Spectra

The central differential operators in orthogonal calculus are the derivatives of a functor. Mimicking the classical notion of a derivative as a measure of infinitesimal change, the first derivative measures how a functor changes when the vector space is extended by a single new direction.

Definition 3.1 First Derivative. For a functor $F \in \mathcal{E}_0$ and an inner-product space V , the first derivative $F^{(1)}$ is the homotopy fiber

$$F^{(1)}(V) = \text{hofiber}[F(V) \xrightarrow{F(\iota)} F(\mathbb{R} \oplus V)],$$

where $\iota : V \hookrightarrow \mathbb{R} \oplus V$ is the standard inclusion. Geometrically, $F^{(1)}(V)$ captures the first-order deviation of F from being constant with respect to dimension.

A crucial feature is that this construction naturally enriches the functor's structure.

property 3.1 Enriched Domain of the Derivative. While F is a functor on $\mathcal{J}_0$, its first derivative $F^{(1)}$ naturally lifts to a functor on the enhanced category $\mathcal{J}_1$.

construction 3.1. [Suspension Map σ] Each derivative functor $F^{(n)}$ is equipped with a natural transformation

$$\sigma : (nV)^c \wedge F^{(n)}(W) \longrightarrow F^{(n)}(V \oplus W).$$

Here, $(nV)^c$ denotes the one-point compactification of $nV = \mathbb{R}^n \otimes V$. This map σ , which is associative and unital in an appropriate sense, serves as a generalized suspension map.

construction 3.2 Higher Derivatives. The process of taking derivatives iterates. Given the n -th derivative $F^{(n)}$, the $(n+1)$ -st derivative is defined inductively as the homotopy fiber

$$F^{(n+1)}(V) = \text{hofiber}[F^{(n)}(V) \rightarrow \Omega^n F^{(n)}(\mathbb{R} \oplus V)].$$

Here, the map is the adjoint of a structure map, and Ω^n denotes the n -fold loop space. This yields a sequence of functors $F^{(n)}$, each a functor on $\mathcal{J}_n$, with $F^{(n+1)}$ measuring the rate of change of $F^{(n)}$.

Each derivative functor comes equipped with an additional pivotal structure.

Definition 3.2 Derivative-at-Infinity Spectrum. The derivative-at-infinity spectrum $\Theta F^{(n)}$ is the spectrum whose k -th space (for k a multiple of n) is $F^{(n)}(\mathbb{R}^k)$, and whose structure maps

$$\Sigma^n F^{(n)}(\mathbb{R}^k) \rightarrow F^{(n)}(\mathbb{R}^{k+1})$$

are specializations of the map σ . This spectrum $\Theta F^{(n)}$ encapsulates the stable, high-dimensional behavior of the n -th derivative.

This structure allows us to assemble the individual values of the derivative into a spectrum.

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A fundamental result is that this spectrum carries a canonical group action.

property 3.2 Orthogonal Group Action on the Derivative Spectrum. The spectrum $\Theta F^{(n)}$ carries a canonical action of the orthogonal group $O(n)$. This action originates from the natural action of $O(n)$ on the category $\mathcal{J}_n$ (by acting on the $\mathbb{R}^n$ factor in the bundle γ_n) and is transported to the derivative via the functorial construction.

Thus, the central algebraic invariant of a functor F in orthogonal calculus is a sequence of spectra $\Theta F^{(1)}, \Theta F^{(2)}, \dots$, each with an action of the corresponding orthogonal group.

4 Taylor Approximation and the Taylor Tower

The orthogonal calculus constructs a sequence of polynomial approximations for a given functor, forming its Taylor tower.

Theorem 4.1 Approximation Functor T_n (Theorem 6.3). For any functor $F \in \mathcal{E}_0$, there exists a natural transformation $\eta_n : F \rightarrow T_n F$, where $T_n F$ is polynomial of degree $\leq n$. It is constructed as

$$T_n F = \text{hocolim}_k (\tau_n)^k F,$$

where τ_n is a certain endofunctor defined via a directional homotopy limit over subspaces of $\mathbb{R}^{n+1}$.

This approximation is universal, meaning $T_n F$ is the best polynomial approximation of degree $\leq n$ to the functor F .

Definition 4.1 Taylor Tower. The approximations assemble into the Taylor tower of F , which is a tower of functors and natural transformations:

$$\cdots \rightarrow T_{n+1} F \xrightarrow{r_{n+1}} T_n F \xrightarrow{r_n} T_{n-1} F \rightarrow \cdots \rightarrow T_0 F.$$

For each V , this yields a tower of spaces $\cdots \rightarrow T_{n+1} F(V) \rightarrow T_n F(V) \rightarrow T_{n-1} F(V) \rightarrow \cdots$.

The layers of this tower, which represent the new information captured at each polynomial degree, are described by a fundamental classification result.

Theorem 4.2 Layer Decomposition (Theorem 9.1). The n -th homotopy fiber (or layer) of the Taylor tower is determined by the n -th derivative spectrum. For an inner-product space V , there is a natural equivalence

$$\mathrm{hofiber}(T_n F(V) \rightarrow T_{n-1} F(V)) \simeq \Omega^\infty \left[((nV)^c \wedge \Theta F^{(n)})_{hO(n)} \right].$$

Here, $(nV)^c$ is the one-point compactification of $\mathbb{R}^n \otimes V$, and $(-)_hO(n)$ denotes the homotopy orbit construction for the canonical $O(n)$ -action on the spectrum $\Theta F^{(n)}$. This shows that the n -th layer is controlled by the Borel construction of the derivative spectrum twisted by V .

This theorem completes the core structure of orthogonal calculus: a functor F is analyzed via its Taylor tower, whose successive layers are homogeneous functors classified by the derivative spectra $\Theta F^{(n)}$ with their orthogonal group actions.

5 Classification of Homogeneous Functors

Within the Taylor tower of a functor F , the layers – the homotopy fibers $\mathrm{hofiber}(T_n F \rightarrow T_{n-1} F)$ – are of a particularly simple and fundamental type: they are homogeneous functors.

Definition 5.1 Homogeneous Functor (Definition 7.1). A functor $D \in \mathcal{E}_0$ is homogeneous of degree n if it satisfies two conditions:

- i) It is polynomial of degree $\leq n$.
- ii) Its $(n-1)$ -st Taylor approximation vanishes: $T_{n-1} D \simeq *$.

Intuitively, a homogeneous functor of degree n is a "pure n -polynomial" object. It contains information precisely of order n and nothing of lower order.

The cornerstone of the theory is the classification theorem for homogeneous functors, which establishes a complete correspondence between algebraic and functorial data.

Theorem 5.1 Classification Theorem for Homogeneous Functors (Theorem 7.3). Let $n > 0$. There is an equivalence of homotopy theories between:

- The category of homogeneous functors of degree n (with morphisms up to equivalence).
- The category of spectra with an action of the orthogonal group $O(n)$.

The correspondence is explicit and constructive.

property 5.1 From Homogeneous Functor to Spectrum. Given a homogeneous functor D , its associated spectrum is precisely its n -th derivative-at-infinity $\Theta D^{(n)}$, equipped with the canonical $O(n)$ -action.

property 5.2 From Spectrum to Homogeneous Functor (Canonical Model). Conversely, given a spectrum Θ with an $O(n)$ -action, the canonical model for a homogeneous functor of degree n is given by:

$$D_{\Theta}(V) \simeq \Omega^{\infty}[(nV)^c \wedge \Theta]_{hO(n)}.$$

The functor D_{Θ} is homogeneous of degree n , and any homogeneous functor D is naturally equivalent to $D_{\Theta D(n)}$.

Remark 1 Comparison with Goodwillie Calculus. This result is the direct analogue of the classification of homogeneous layers in Goodwillie's calculus of functors from spaces to spaces. However, a critical difference lies in the symmetry group involved:

- In Goodwillie calculus, the n -th homogeneous layer is classified by a spectrum with an action of the symmetric group Σ_n . This reflects that the n -th cross-effect deals with an n -fold product of spaces, on which Σ_n acts by permuting factors.
- In orthogonal calculus, the n -th derivative involves the vector bundle γ_n over the Stiefel manifold, whose fiber is $\mathbb{R}^n \otimes \text{coker}(f)$. The group that naturally acts on the $\mathbb{R}^n$ tensor factor is the $O(n)$. This geometric origin explains why $O(n)$ -actions govern the classification in this setting.

This distinction is the hallmark of orthogonal symmetry.

Thus, the classification theorem beautifully synthesizes the calculus: the n -th layer of the Taylor tower of any functor F is completely determined by the spectrum $\Theta F^{(n)}$, and this data is packaged with the canonical linear symmetry group $O(n)$.

6 Example

Example 6.1 Derivatives of $BO(V)$ (Example 2.7, 10.3). The functor $F(V) = BO(V)$ is a paradigmatic example. Its Taylor tower encodes and generalizes classical characteristic class theory.

- First Derivative: $\Theta F^{(1)} \simeq \mathbb{S}$, the sphere spectrum, with the trivial action of $O(1)$. This reflects that the first-order change in $BO(V)$ when adding a line is related to the space of linear frames. The associated tower layer recovers information about Stiefel-Whitney classes.
- Second Derivative: $\Theta F^{(2)} \simeq \mathbb{S}[-1]$. Weiss interprets this stage as capturing the theory of Pontryagin classes in a generalized, intrinsic form.
- Third Derivative: $\Theta F^{(3)} \simeq \Omega^2 \mathbf{mo}(3)$, a double desuspension of the mod-3 Moore spectrum. This nontrivial result shows the emergence of torsion phenomena in higher-order approximations.

A key observation is that the map $\eta_2 : BO(V) \rightarrow T_2 BO(V)$ is a rational homotopy equivalence for $V \neq 0$, meaning the quadratic Taylor approximation captures all rational cohomological information of $BO(V)$.

Example 6.2 Derivatives of $B\text{Top}(V)$. For the functor $G(V) = B\text{Top}(V)$, which classifies microbundles with fiber V , the first derivative has a profound interpretation: $\Theta G^{(1)} \simeq \mathbf{A}(*).$ This is Waldhausen's A -theory spectrum of a point, a cornerstone of algebraic K -theory of spaces.

This result, due to Weiss and Williams, creates a direct bridge between calculus methods and geometric topology, identifying the primary "rate of change" of topological manifold theory with an algebraic K -theory invariant. The higher derivatives of this functor are largely unknown and remain a fascinating area of study.

Example 6.3 Application to Embedding Spaces $\text{Emb}(M, \mathbb{R}^\infty)$. Perhaps the most significant application is to the study of embedding spaces, as developed in [ALV07]. For a compact manifold M , the functor $V \mapsto \text{Emb}(M, V)$ is an object of orthogonal calculus. Its Taylor tower provides a filtration of the space $\text{Emb}(M, \mathbb{R}^\infty)$.

A key result is that when $\dim V \geq 2 \dim M + 2$, the rational homotopy type of $\text{Emb}(M, V)$ is completely determined by this tower. Furthermore, the rational homotopy groups of the layers—the homogeneous functors—can be explicitly calculated using graph complexes. These are combinatorial chain complexes whose homology encodes data about configurations of points in M and their linking, demonstrating how orthogonal calculus translates a geometric topology problem into a computable algebraic one.

7 Relation to Goodwillie Calculus

Orthogonal calculus and Goodwillie calculus are sister theories, sharing the fundamental philosophy of approximating functors by polynomial ones in a homotopy-theoretic sense. However, they are designed for different settings, leading to crucial technical and conceptual distinctions.

- property 7.1 Primary Difference: Domain.** • Goodwillie calculus operates on functors whose domain is a category of spaces or spectra (e.g., $F : \mathcal{S} \rightarrow \mathcal{S}$). Its constructions are intrinsic to homotopy theory, relying on interactions between products and coproducts.
- Orthogonal calculus is tailor-made for functors defined on the category $\mathcal{J}$ of Euclidean vector spaces and isometries (e.g., $F : \mathcal{J} \rightarrow \mathcal{S}$). This domain carries a rich linear structure, which the calculus exploits.

This foundational difference dictates the algebraic structure of the derivatives.

- property 7.2 Symmetry Groups in Classification.** • In Goodwillie calculus, the n -th derivative is a spectrum with an action of the symmetric group Σ_n . This arises because the n -th cross-effect involves an n -fold smash product or cartesian product, on which permutations act naturally.
- In orthogonal calculus, derivatives are spectra with an action of the orthogonal group $O(n)$. This action originates from the $O(n)$ -action on the vector bundle $\gamma_n(V, W) = \mathbb{R}^n \otimes \text{coker}(f)$ used to define the higher morphism categories $\mathcal{J}_n$.

Consequently, their primary fields of application are distinct.